

GIRISH SOMAN S G

Senior Software Engineer (Node.js | TypeScript | AWS)

Kerala, India | giri943@gmail.com | +91-9446114870 | +91-7012204262 | [LinkedIn](#)

Professional Summary

Software Engineer with 5+ years of hands-on experience designing, scaling, and maintaining distributed backend systems, event-driven messaging microservices, and observability platforms using Node.js, TypeScript, and AWS. Proven track record of developing real-time monitoring solutions, AI-driven WhatsApp integration, queue-based background processing (AWS SQS), and high-availability enterprise services. Adept at unit/integration testing (Jest), database query optimization (PostgreSQL/MongoDB), and engineering fault-tolerant, SLA-compliant cloud infrastructure.

Areas of Expertise

• TypeScript	• JavaScript (ES6+)	• Node.js	• SQL
• NoSQL	• AWS (EC2, S3, CloudFront, RDS, SQS, SES, SNS, Rekognition)	• GCP	• GitHub
• Docker	• CI/CD	• REST APIs	• Next.js
• Strapi	• WordPress	• Postman	• Jest.js
• Claude Code	• Cursor	• Kiro	• Co Pilot

Accomplishments

- Architected and launched Schbang Pulse — a real-time website/API monitoring and automated alerting platform using TypeScript, Node.js, Server-Sent Events and AWS, delivering production-grade observability and incident notification.
- Engineered an AI-driven WhatsApp chatbot backend with asynchronous processing (AWS SQS) and email workflows (AWS SES), enabling reliable handling of high-volume webhooks and automated messaging pipelines.
- Optimized database schemas and queries (PostgreSQL, MySQL, MongoDB), cutting peak API response times by up to 40% and improving overall system throughput.
- Designed and implemented secure, scalable REST APIs with JWT/Google OAuth authentication, strict rate-limiting, and Role-Based Access Control to ensure compliance and high availability.
- Integrated third-party AI and messaging services (OpenAI, Replicate, Meta/WhatsApp, Google Chat) to build multimodal conversational interfaces and multi-channel incident alerting.
- Built robust background job processing and retry strategies using queue-based architecture (SQS), reducing failed job impact and improving SLA adherence.
- Improved release quality through unit and integration testing (Jest, Postman) and CI/CD pipelines, increasing deployment confidence and reducing production regressions.
- Implemented cloud-native delivery patterns (AWS S3, CloudFront) for high-volume static asset distribution and edge optimization to ensure reliable content delivery.

Professional Experience

Senior Software Engineer, Schbang Digital Solutions, Mumbai, India (Remote)

Feb 2021 to Present

- **Distributed Systems & Observability:** Architected real-time website/API monitoring, uptime tracking, and automated alerting services (Schbang Pulse) utilizing TypeScript, Node.js, and Server-Sent Events (SSE).

- **Asynchronous Queue & Chatbot Engineering:** Engineered an AI-driven WhatsApp chatbot backend utilizing Node.js, AWS SQS, and AWS SES, processing high volumes of incoming webhooks, asynchronous jobs, and automated messaging workflows.
- **API & System Architecture:** Built secure, scalable RESTful APIs with strict rate-limiting, JWT/Google OAuth authentication, and Role-Based Access Control (RBAC), ensuring high availability and system resilience.
- **Third-Party Channel Integrations:** Integrated external communication and media APIs (Meta, WhatsApp, Google Chat API, OpenAI, Replicate AI) to build conversational AI interfaces and multi-channel incident alerting pipelines.
- **Database & Performance Optimization:** Optimized SQL (PostgreSQL/MySQL) and NoSQL (MongoDB) schema indexing and aggregation queries, cutting peak API response times by up to 40%.
- **Cloud Infrastructure:** Utilized AWS S3 and CloudFront for high-volume static asset delivery and edge delivery optimization, ensuring reliable content distribution.
- **Code Quality & Automated Testing:** Implemented unit tests using Jest and automated testing suites via Postman to validate API functionality, error handling, and system reliability across release cycles.
- **Agile Collaboration:** Worked closely with cross-functional DevOps, Frontend, QA, and Product teams using Git/GitHub workflows and Jira/Asana in fast-paced Agile environments.

Education

University of Madras

Bachelor of Business Administration (BBA), 2016 to 2019

Links

[Linkedin](#), [Github](#) , [Portfolio](#)